

Homework 2

*Instructor: Henry Lam**Due: Sept 26, 2025*

Problem 1 Every evening, two weather stations issue weather forecast for the next day. The weather forecasts are independent. On average, the weather forecast of station 1 is correct 90% of the cases, irrespective of weather type. This percentage is 80% for station 2. Suppose that, without any forecast, the belief is that the next day rains or has sunny weather each with a 50% chance. On a given day, station 1 predicts sunny weather for the next day, whereas station 2 predicts rain. What is the probability that the weather forecast of station 1 will be correct?

Problem 2 *USA Today* reported that “high-volume” hospitals performed 77% of bladder removal surgeries; “low-volume” hospitals performed 23%. In the first two weeks after surgery, 3.1% of patients at low-volume centers died, compared to 0.7% at the high-volume centers.

- Compute the probability that a randomly chosen bladder-cancer patient had surgery at a high-volume hospital and survived the first two weeks after surgery.
- Calculate the probability that a randomly chosen bladder-cancer patient survived the first two weeks after surgery.
- If two bladder-cancer patients were chosen randomly, determine the probability that only one would survive the first two weeks after surgery.
- If three bladder-cancer patients were chosen randomly, determine the probability that at least two would survive the first two weeks after surgery.

Problem 3 This year experts project that 20% of all taxpayers will file an incorrect tax return. The Internal Revenue Service (IRS) itself is not perfect. IRS auditors claim that there is an error when no problem exists about 10% of the time. The audits also indicate no error with a tax return when in fact there really is a problem about 30% of the time.

The IRS has just notified a taxpayer there is an error in his return. What is the probability that the return actually has an error?

Problem 4. On independent events:

- Consider two independent events A and B. Prove that A and B^c are also independent.
- Prove that if $E_1, E_2, \dots, E_n$ are independent events, then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = 1 - \prod_{i=1}^n [1 - P(E_i)]$$

Problem 5. Alice is hired by CVS to improve its overall product display optimization in its New York stores. For some reason, she begins her work with flashlights in the electrical supplies section. She finds out that in the display of electrical supplies section in each store in New York, 20% of the flashlights are of type 1, 30% are of type 2 and 50% are of type 3. In addition, she learns that the probability that Type 1 flashlight will give over 100 hours of use is 0.7, with the corresponding probabilities for Type 2 and Type 3 flashlights being 0.4 and 0.3 respectively.

- To model customer buying behaviour, Alice assumes that any customer who wants to buy a flashlight chooses one uniformly at random from the display. Under this assumption, what is the probability that a randomly chosen flashlight that a customer buys will give more than 100 hours of use?
- Given the flashlight lasted over 100 hours, what is the conditional probability that it was a type j flashlight, with $j = 1, 2, 3$

Problem 6. A pair of fair dice is rolled. Let E denote the event that the sum of the dice is equal to 7

- (a) Show that E is independent of the event that the first die lands on 4
- (b) Show that E is independent of the event that the second die lands on 3 .

Problem 7. Let A, B, C be events such that $P(A) = 0.2, P(B) = 0.3, P(C) = 0.4$ Find the probability that at least one of the events A and B occurs if

- (a) A and B are mutually exclusive;
- (b) A and B are independent.

Find the probability that all of the events A, B, C occurs if

- (c) A, B, C are independent;
- (d) A, B, C are mutually exclusive.